Vehicle Parameter Identification Report

1 Vehicle Parameter Identification

The dynamic bicycle model used by the MPC controller requires physical parameters that cannot be taken from datasheets alone. This section describes the experimental procedures used to identify them on the real F1/10th vehicle (Traxxas Slash 4×4 chassis, VESC 6 controller, Kv = 3500 motor).

All tests use a common ROS 2 framework that records time-stamped sensor data at approximately 200 Hz: VESC odometry (speed from ERPM), the onboard IMU (3-axis accelerometer and gyroscope), and motor state (current, voltage). Each test is run five times; the file with the highest sample count is used for reporting. Raw data is saved to CSV for post-processing.

Quantity	**Value**	**Source**
Total mass m	3.314 kg	Measured (scale)
Wheelbase L	0.324 m	CAD model
CG to front axle l_f	0.166 m	CAD model
CG to rear axle l_r	0.160 m	CAD model
Effective wheel radius r_{eff}	0.051 m	Measured
Gear ratio G	11.82	Datasheet
Motor K_v	3500 RPM/V	Datasheet

Vehicle constants (measured / CAD).

1.1 Dynamic Bicycle Model

The MPC uses a 7-state dynamic bicycle model with state $\mathbf{x} = [x,\, y,\, \psi,\, v_x,\, v_y,\, \omega,\, \omega_w]$ and control inputs $\mathbf{u} = [\delta,\, T_{\text{motor}}]$. The equations of motion are:

$$\dot{v}_x = \frac{1}{m}\big(F_x - F_{yf}\sin\delta + m\, v_y\, \omega\big) \tag{1}$$

$$\dot{v}_y = \frac{1}{m}\big(F_{yf}\cos\delta + F_{yr} - m\, v_x\, \omega\big) \tag{2}$$

$$\dot{\omega} = \frac{1}{I_z}\big(l_f\, F_{yf}\cos\delta - l_r\, F_{yr}\big) \tag{3}$$

$$\dot{\omega}_w = \frac{1}{I_w}\Big(\frac{T_{\text{motor}}}{G} - F_x\, R_w\Big) \tag{4}$$

where G is the gear ratio, R_w the wheel radius, and I_w the drivetrain inertia. Lateral tire forces use a linear model:

$$F_{yf} = C_{\alpha f}\, \alpha_f, \qquad F_{yr} = C_{\alpha r}\, \alpha_r \tag{5}$$

with slip angles

$$\alpha_f = \delta - \arctan\frac{v_y + l_f\, \omega}{v_x}, \qquad \alpha_r = -\arctan\frac{v_y - l_r\, \omega}{v_x}. \tag{6}$$

The longitudinal force is modelled as $F_x = m\,a_{\mathrm{cmd}}$ (the MPC commands acceleration directly via the VESC). Table 6 lists all model parameters and the test used to identify each one.

1.2 Tire Friction Coefficient

Theory. At steady-state circular driving the centripetal acceleration is $a_y = v^2/R$. The maximum sustainable a_y before the tires begin to slide gives the friction coefficient:

$$\mu = \frac{a_{y,\max}}{g}. \tag{7}$$

The IMU measures a_y directly, independent of odometry errors.

Procedure. The car drives circles at a fixed steering angle ($\delta_{\mathrm{cmd}} = 0.3\,\mathrm{rad}$) with speed increasing in $0.5\,\mathrm{m\,s^{-1}}$ steps from $2.0\,\mathrm{m\,s^{-1}}$ to $5.0\,\mathrm{m\,s^{-1}}$. At each speed, 8 s of steady-state IMU data is recorded.

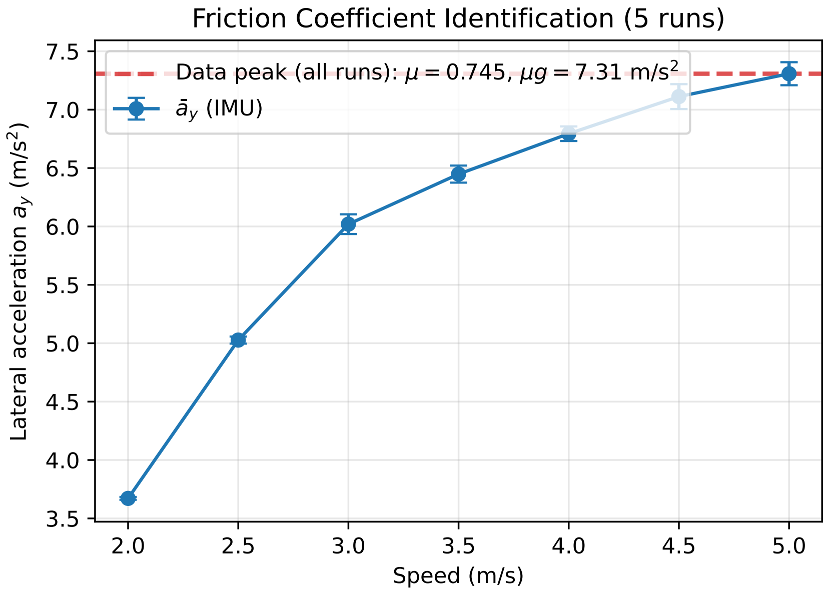

Figure 1: Lateral acceleration (IMU) vs. speed, grouped by speed step. The friction coefficient μ is determined from the saturation plateau at high speed.

Table 1: Friction test results: mean lateral acceleration and friction coefficient at each speed step (5 runs, ∼32 000 samples per speed).

Speed (m/s)	$\bar{a}_y$ (m/s^2)	$\mu = a_y/g$
2.0	3.67 ± 0.01	0.374
2.5	5.03 ± 0.03	0.512
3.0	6.02 ± 0.08	0.614
3.5	6.45 ± 0.07	0.657
4.0	6.79 ± 0.06	0.693
4.5	7.11 ± 0.11	0.725
5.0	7.31 ± 0.10	0.745

Data. The measured μ increases with speed and begins to plateau at $v \geq 4.5\,\mathrm{m\,s^{-1}}$, approaching $\mu \approx 0.75$. This value is consistent with rubber tires on smooth indoor flooring. The increasing standard deviation at higher speeds reflects the onset of intermittent sliding as the traction limit is approached.

Validation. The friction coefficient is cross-checked against the motor torque test (Section 1.5), where the traction force saturates at $F_{\text{sat}} \approx 24\,\text{N}$, giving $\mu_x = F_{\text{sat}}/(m\,g) = 24/(3.314 \times 9.81) \approx 0.74$. Both estimates are consistent, confirming $\mu \approx 0.74$–0.75 depending on the loading direction (longitudinal vs. lateral).

Uncertainty.

- IMU bias: 2 s stationary calibration; residual $< 0.05\,\text{m}\,\text{s}^{-2}$.
- Result is surface-specific (smooth indoor floor); μ will differ on carpet, concrete, or asphalt.
- Tire temperature affects grip; all tests run at ambient temperature.

1.3 Cornering Stiffness

Theory. In steady-state cornering ($\dot{v}_y = \dot{\omega} = 0$), the lateral force balance gives front and rear tyre forces

$$F_{yf} = m\,a_y\,\frac{l_r}{L}, \qquad F_{yr} = m\,a_y\,\frac{l_f}{L}, \tag{8}$$

and the linearised slip angles are

$$\alpha_f = \beta + \frac{l_f\,\omega}{v_x} - \delta, \qquad \alpha_r = \beta - \frac{l_r\,\omega}{v_x}, \tag{9}$$

where $\beta = v_y/v_x$ is the body sideslip angle. The cornering stiffnesses follow from $C_{\alpha f} = -F_{yf}/\alpha_f$ and $C_{\alpha r} = -F_{yr}/\alpha_r$ in the assumed linear tyre region.

Because the VESC odometry reports $v_y = 0$ and no sensor on this platform provides a reliable lateral-velocity estimate, the per-point slip angle method cannot separate tyre properties from kinematics. The understeer gradient method avoids this limitation entirely.

Understeer gradient method. The understeer gradient K_{us} is a well-conditioned observable that characterises the vehicle's steering balance without requiring v_y. In steady-state circular driving:

$$\delta - \frac{L\,\omega}{v_x} = K_{\text{us}}\,a_y, \qquad K_{\text{us}} = \frac{m}{L}\left(\frac{l_f}{C_{\alpha r}} - \frac{l_r}{C_{\alpha f}}\right). \tag{10}$$

The quantity $\delta - L\omega/v_x$ is the deviation of the measured steering from the kinematic (Ackermann) steering angle; it equals zero for a neutral car and is positive for understeer.

Procedure. Steady-state circles were driven at five commanded steering angles ($\delta_{\text{cmd}} = 0.12$, 0.16, 0.20, 0.24, 0.28 rad) and four speeds (1.5, 2.0, 2.5, $3.0\,\text{m}\,\text{s}^{-1}$), giving 20 operating points per run. Each point was held for 10 s after a 4 s settling period, recording v_x, ω, a_y, and δ at approximately 200 Hz. The full test was repeated five times, yielding 100 data points in total.

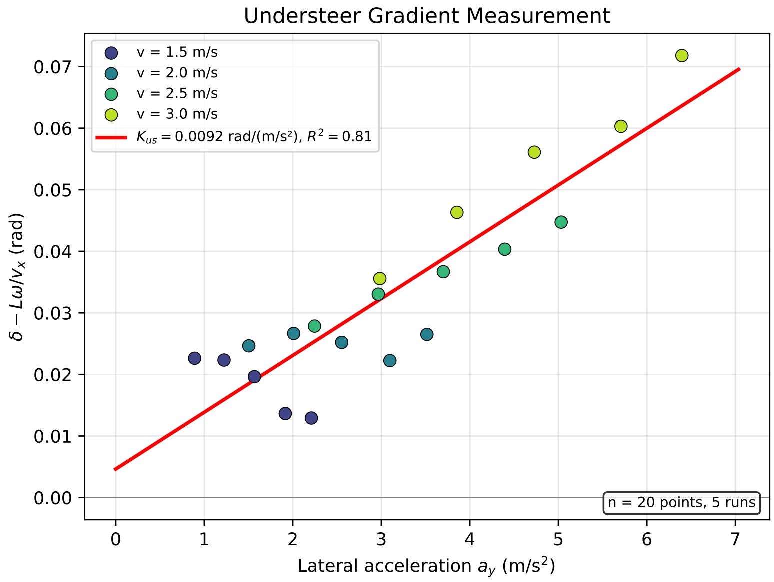

Figure 2: Understeer gradient measurement. The vertical axis is $\delta - L\omega/v_x$ (deviation from kinematic steering); the horizontal axis is the lateral acceleration a_y. The slope of the linear fit gives $K_{\mathrm{us}} = 0.0092\,\mathrm{rad/(m/s^2)}$ ($R^2 = 0.81$, $n = 20$ operating points, 5 runs).

Results. The steering angle used in the regression is the calibrated wheel angle from the VESC servo mapping (Section 1.8), corrected for the nonlinear servo response (Section 1.9). The measured understeer gradient is $K_{\mathrm{us}} = 0.0092 \pm 0.001$ rad/(m/s^2). The positive value confirms the vehicle understeers, requiring progressively more steering input at higher lateral accelerations.

Individual cornering stiffness. The understeer gradient (10) provides one equation relating two unknowns ($C_{\alpha f}, C_{\alpha r}$). A second constraint is obtained from the yaw-plane natural frequency of the single-track model:

$$\omega_n^2 = \frac{1}{I_z}\left(l_f^2\, C_{\alpha f} + l_r^2\, C_{\alpha r}\right) + \frac{C_{\alpha f} + C_{\alpha r}}{m} - v_x^2. \tag{11}$$

For a given ω_n, the system (10)–(11) can be solved for the individual stiffnesses. Figure 3 shows the parametric solution.

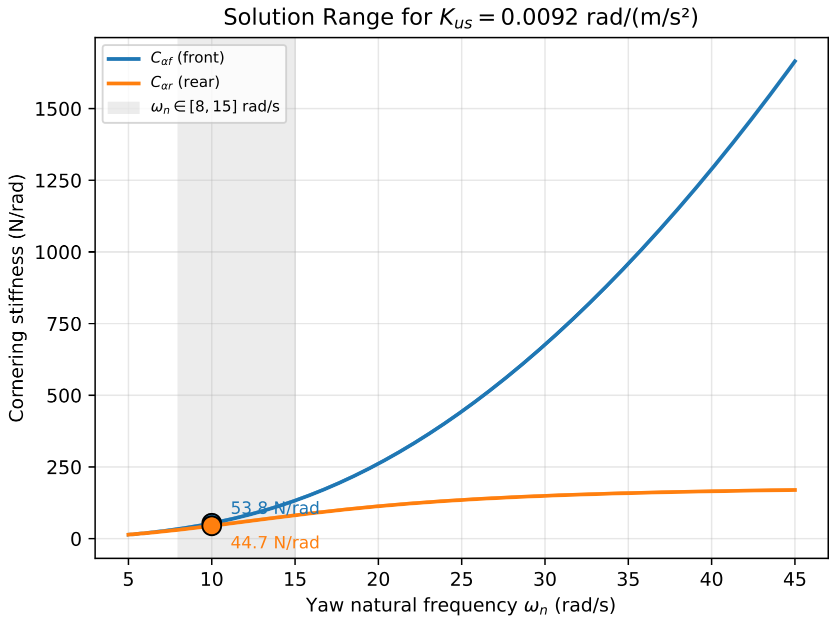

Figure 3: Cornering stiffness pairs $(C_{\alpha f}, C_{\alpha r})$ satisfying the measured $K_{\text{us}} = 0.0092$ as a function of the assumed yaw-plane natural frequency ω_n. The shaded band marks an estimated frequency range.

At $\omega_n = 10\,\text{rad}\,\text{s}^{-1}$, the solution is:

$$C_{\alpha f} = 53.6\,\text{N}\,\text{rad}^{-1}, \qquad C_{\alpha r} = 44.6\,\text{N}\,\text{rad}^{-1}. \tag{12}$$

The corresponding normalised stiffnesses for the dynamic bicycle model are $C_{Sf} = C_{\alpha f}/(m\,g) = 1.65$ and $C_{Sr} = C_{\alpha r}/(m\,g) = 1.37$. The absolute values depend on the assumed ω_n; the ratio $C_{\alpha f}/C_{\alpha r} \approx 1.20$ is less sensitive and consistent with the measured understeer behaviour.

Can the Pacejka shape factor C be identified? Not uniquely from the present dataset. The experiments provide (i) near-linear slip-angle behaviour through K_{us} and the derived C_α values, and (ii) a friction limit μ from peak lateral acceleration. In the simplified Magic Formula $F_y = D\sin\big(C\arctan(B\alpha)\big)$, these constrain $D \approx \mu F_z$ and the low-slip slope $BCD \approx C_\alpha$, but do not separate B and C. Therefore C must be assumed or bounded from prior tire data unless additional nonlinear $F_y(\alpha)$ sweep measurements near saturation are collected.

1.4 Rolling Resistance

Theory. When the throttle is released and the motor coasts freely, the only decelerating forces are rolling resistance and aerodynamic drag:

$$m\,a_x = -c_{\text{roll}}\,m\,g - \tfrac{1}{2}\rho\,C_d\,A\,v^2.$$

At F1/10th scale and $v < 4\,\text{m}\,\text{s}^{-1}$, aerodynamic drag is negligible ($< 0.03\,\text{N}$), giving $c_{\text{roll}} \approx |a_x|/g$.

Procedure. The car accelerates to $3\,\text{m}\,\text{s}^{-1}$, then the motor current is set to zero (bypassing the VESC speed controller, which would actively brake) and the car coasts to a stop. Ten runs are performed. The deceleration is measured from wheel odometry (ERPM), which is accurate during free rolling (no traction slip).

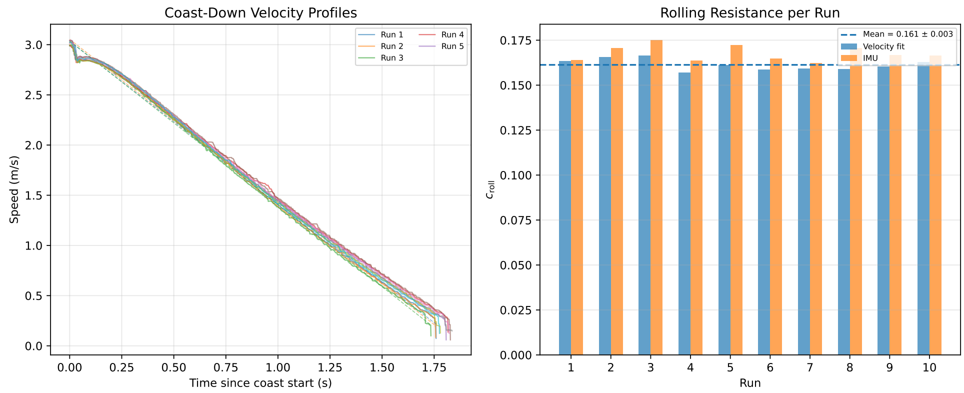

Figure 4: Left: coast-down velocity profiles for all ten runs. Right: per-run c_{roll} from both the velocity-fit and IMU methods.

Results. A linear fit of $v(t)$ during the coast phase gives the deceleration a_{coast}, from which $c_{\text{roll}} = |a_{\text{coast}}|/g$. Over ten runs, the velocity-fit method yields $c_{\text{roll}} = 0.163 \pm 0.003$ (IMU method: 0.163 ± 0.004). Mean motor current during coasting was 0.37 A (below the 1 A threshold), confirming the motor was freewheeling. This value includes both tire rolling resistance and drivetrain friction (gear mesh, bearings), which is the total resistive coefficient needed by the minimum-time optimizer.

1.5 Motor Torque Mapping

Theory. Two independent estimates of wheel force:

$$F_{\text{current}} = \frac{K_t \cdot G}{r_{\text{eff}}} I_{\text{motor}}, \tag{13}$$

$$F_{\text{accel}} = m\, a_x \quad \text{(IMU)}, \tag{14}$$

where $K_t = 60/(2\pi\, K_v) = 0.002\,73\,\mathrm{N\,m\,A^{-1}}$ and $G = 11.82$. Plotting F_{accel} against I_{motor} yields the effective current-to-force gain (slope) and allows identification of the VESC current limit.

Procedure. Accelerate from rest to target speeds (1.5, 2.0, 2.5, $3.0\,\mathrm{m\,s^{-1}}$), recording motor current and IMU a_x simultaneously.

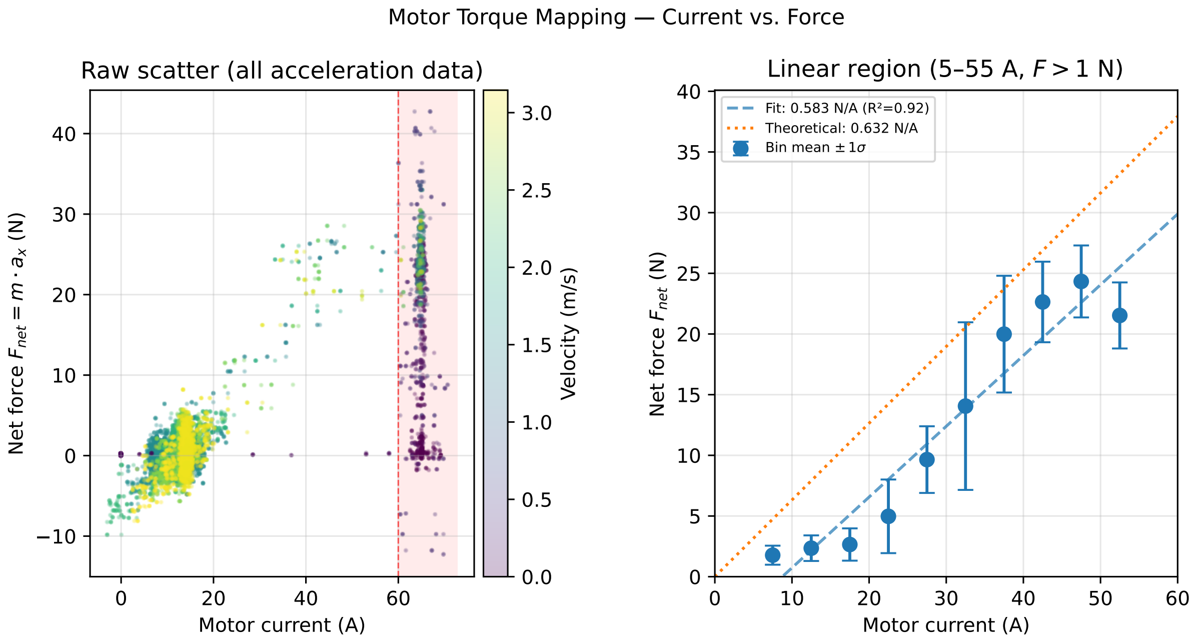

Figure 5: Motor torque mapping. *Left:* raw scatter of all acceleration-phase data, coloured by velocity. The red shaded region marks the VESC current limit (∼65 A). *Right:* bin-averaged force vs. current in the linear region (5–55 A), with the measured fit and theoretical line.

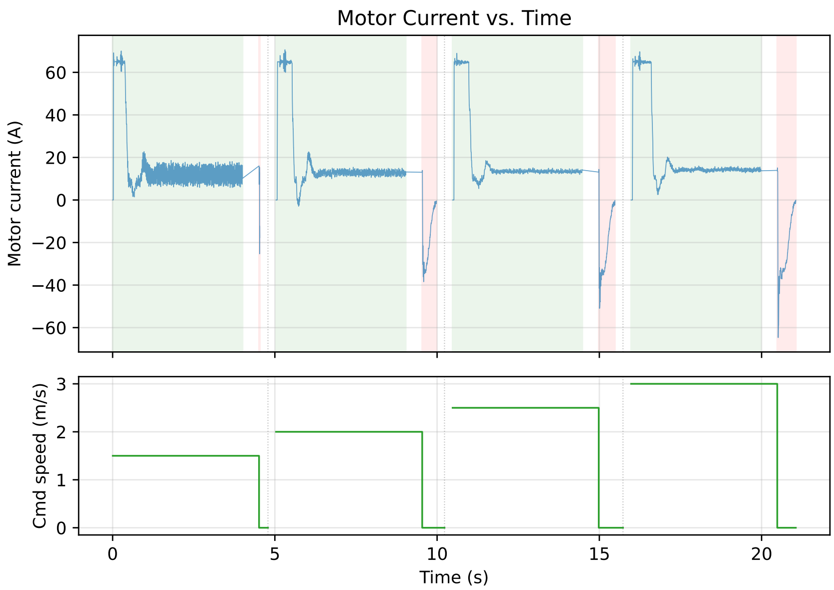

Figure 6: Motor current (top) and commanded speed (bottom) vs. time for each speed step. Green/red shading indicates acceleration/braking phases. Time gaps between steps are collapsed.

Table 2: Motor torque test: per-speed current and force statistics during the acceleration phase.

Cmd speed (m/s)	$\bar{I}$ (A)	Peak I (A)	$\bar{F}_{\text{net}}$ (N)	Samples
1.5	16.9	70.0	1.5	3204
2.0	18.7	70.6	2.0	3236
2.5	19.6	69.0	2.4	3235
3.0	21.5	69.7	2.8	3233

Data. The mean current and force increase with speed as expected. The peak current of $\sim$70 A at every speed step corresponds to the VESC current limit during the initial launch from standstill. The mean force is low because the average includes the steady-state cruising period where the car has reached its target speed and acceleration is near zero.

Key results.

- **Measured** $K_{t,\text{eff}}$: $0.583\,\text{N}\,\text{A}^{-1}$ (bin-averaged fit, $R^2 = 0.92$).
- **Theoretical** $K_t\,G/r_{\text{eff}}$: $0.632\,\text{N}\,\text{A}^{-1}$.
- **Efficiency**: $\eta = 0.583/0.632 = 92\%$. The 8% loss is attributed to gearbox friction, belt compliance, and bearing drag — consistent with hobby-grade drivetrains.
- **VESC current limit**: $\sim$65 A, producing a maximum wheel force of $F_{\max} = 0.583 \times 65 = 37.9\,\text{N}$ and a corresponding wheel torque of $T_{\text{drive}} = F_{\max} \cdot r_{\text{eff}} = 1.93\,\text{N}\,\text{m}$.

Uncertainty.

- Motor K_v from datasheet ($\pm 5\%$ typical for hobby motors).
- r_{eff} measurement accuracy ($\pm 1\,\text{mm} \Rightarrow\sim 2\%$ error on force).
- Gear efficiency not measured separately; absorbed into effective $K_{t,\text{eff}}$.

VESC current limiting. During the motor torque tests, the motor current was observed to peak at approximately 71.6 A during burst acceleration and −63.5 A during regenerative braking. Based on these observations, the VESC motor current limit was configured to 100 A (motor) and 70 A (battery) in the VESC Tool to ensure the motor and ESC operate within safe thermal limits. Data above the 60 A threshold was excluded from the $K_{t,\text{eff}}$ regression because the VESC current limiter saturates the motor current in this region, producing a non-linear force–current relationship. Below this threshold the force–current relationship is linear and the measured $K_{t,\text{eff}}$ is valid.

1.6 Maximum Dynamics

Theory. Direct measurement of the vehicle's performance envelope: $v_{\max}$, $a_{\max}$ (peak acceleration), and a_{brake} (peak deceleration). These define the MPC input and state constraints.

Procedure. Command maximum speed for 5 s (acceleration phase), then command zero (deceleration phase). Both ERPM-based speed and IMU a_x are recorded.

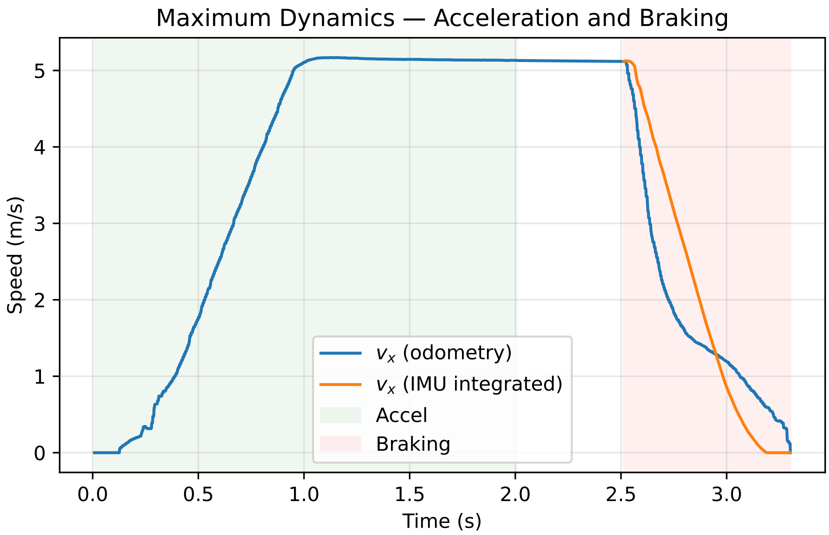

Figure 7: Speed (odometry) vs. time during a maximum dynamics run. The braking region shows odometry dropping to zero quickly as the wheels lock.

Table 3: Maximum dynamics results.

Quantity	**Value**	**Source**
Maximum velocity $v_{\max}$	$5.17\,\mathrm{m\,s^{-1}}$	ERPM odometry
Peak acceleration $a_{\max}$	$7.3\,\mathrm{m\,s^{-2}}$	Friction-bound (μg)
Peak deceleration a_{brake}	$-11.06\,\mathrm{m\,s^{-2}}$	IMU (smoothed)

Data.

Validation. The smoothed IMU acceleration peak can exceed the physical traction limit due to pitch dynamics and sensor coupling. Therefore the longitudinal acceleration used for constraints is conservatively bounded by the friction estimate, $a_{\max} = \mu g \approx 7.3\,\mathrm{m\,s^{-2}}$.

The peak deceleration $|a_{\text{brake}}| = 11.06\,\mathrm{m\,s^{-2}}$ exceeds $\mu\, g \approx 7.3\,\mathrm{m\,s^{-2}}$, which indicates that the IMU reading during braking includes a component from chassis pitch (the accelerometer cannot distinguish deceleration from forward pitch rotation under gravity). For MPC constraints, the conservative value $a_{\text{brake}} = \mu\, g \approx 7.3\,\mathrm{m\,s^{-2}}$ is recommended.

Uncertainty.

- ERPM-derived speed is accurate during acceleration but unreliable during braking (wheel lock/slip).
- IMU-integrated velocity drifts; smoothing introduces lag.
- Battery voltage sag under load reduces effective $v_{\max}$ as the battery discharges.

1.7 Steering Actuator Response

Theory. The steering servo has a finite angular velocity. By commanding step changes from $0 \to \delta$ and measuring the 10%–90% rise time of the IMU yaw rate response:

$$\dot{\delta}_{\max} = \frac{0.8\,\delta_{\text{cmd}}}{\Delta t_{10\text{–}90}}. \tag{15}$$

The factor 0.8 accounts for measuring the 10% to 90% rise region (i.e. 80% of the full step amplitude).

Procedure. Drive at constant $1.5\,\mathrm{m\,s^{-1}}$. Alternate steering between $+0.3$ rad and -0.3 rad for 6 steps (each from the straight-ahead position), with 1.5-second straight pauses between steps. Record IMU yaw rate ω_z and commanded steering angle.

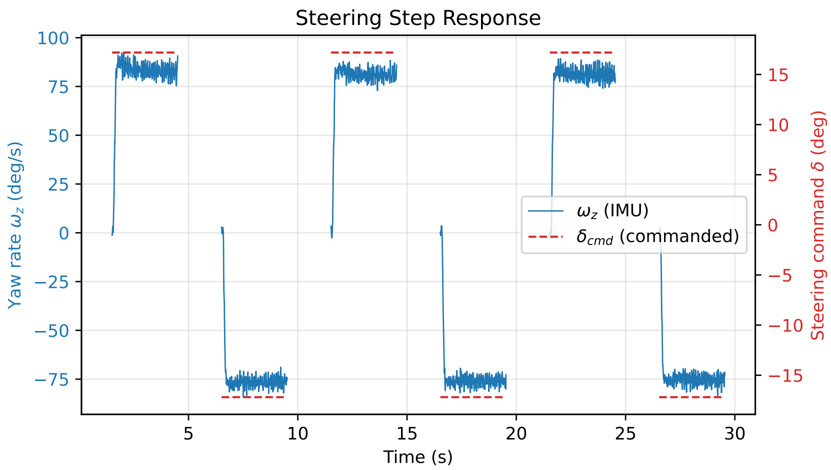

Figure 8: IMU yaw rate (solid blue, left axis) and commanded steering angle (dashed red, right axis) vs. time. Lines are broken at the recording gaps between steps.

Table 4: Steering step response: 10%–90% rise time of the yaw-rate response for each transition.

Step	$\Delta\omega$ (deg/s)	Rise time (ms)	$\dot{\delta}$ (rad/s)
0	84.1	95	2.53
1	−79.2	78	3.08
2	77.9	82	2.93
3	−75.9	76	3.16
4	81.0	92	2.61
5	−75.9	96	2.50
Average		87	2.80[†]

[†]Estimated as $\dot{\delta}_{\max} = 0.8 \times 0.3$ rad / 87 ms $= 2.80\,\mathrm{rad\,s^{-1}}$.

Data. The yaw rate response shows that the combined servo and vehicle dynamics produce consistent 10–90% rise times of ∼87 ms. The estimated maximum steering rate of $2.85\,\mathrm{rad\,s^{-1}}$ is close to the simulation default of $3.2\,\mathrm{rad\,s^{-1}}$.

1.8 VESC Calibration

Procedure. The VESC maps a commanded steering angle to a servo PWM position via $s = g \cdot (-\delta_{\text{cmd}}) + b$. The servo gain $g = -0.7284$, offset $b = 0.5500$, and servo limits (`servo_min` $= 0.202$, `servo_max` $= 0.890$) were determined with `find_servo_limits.py` and `test_steering_calibration.py`. The speed-to-ERPM gain was calibrated to 4550 by comparing VESC-reported velocity against ground-truth distance measurements. Table 5 lists the calibrated values.

Table 5: Calibrated VESC parameters.

Parameter	Value
`speed_to_erpm_gain`	4550
`steering_angle_to_servo_gain`	-0.7284
`steering_angle_to_servo_offset`	0.5500
`servo_min` / `servo_max`	0.202 / 0.890

1.9 Servo Nonlinearity Correction

Motivation. The standard F1/10th VESC driver assumes that the actual wheel angle is proportional to the commanded angle, i.e. $\delta_{\text{act}} = \delta_{\text{cmd}}$, after applying the servo gain and offset. Physical measurements with a digital protractor revealed that the relationship between the servo PWM value s and the actual wheel angle is *nonlinear*, introducing a systematic error that grows with steering magnitude.

Physical servo mapping. The actual wheel angle was measured at 15 servo positions spanning the full range. A second-order polynomial fit to the data gives:

$$\delta_{\text{act}}(s) = -0.9262\, s^2 - 0.4778\, s + 0.5365, \tag{16}$$

where s is the servo value ($s = g \cdot \delta_{\text{cmd}} + b$, with gain g and offset b from Table 5). The zero-steering offset $\delta_{\text{act}}(b)$ is subtracted so that $\delta_{\text{act}} = 0$ when $\delta_{\text{cmd}} = 0$.

Correction polynomial. To compensate the nonlinearity in real time, a correction is applied in the VESC Ackermann node before the standard gain/offset mapping:

$$\delta_{\text{corrected}} = c_2\, |\delta_{\text{cmd}}|^2 + c_1\, |\delta_{\text{cmd}}| + c_0, \tag{17}$$

with $c_2 = 0.5896$, $c_1 = 0.9181$, $c_0 = 0.0015$ (sign restored from δ_{cmd}). The correction was validated by repeating the protractor measurements with the polynomial active: max $|\text{error}| =$ 1.8%, mean = 1.0%, RMS = 1.2%.

Impact on cornering stiffness. The cornering stiffness tests (Section 1.3) were conducted *before* the polynomial correction was deployed, using an earlier servo gain of $g_{\text{old}} = -0.7940$. With this gain, the servo overshoot caused actual wheel angles to exceed the commanded values by up to 11.8% at small angles. Retrospectively correcting the commanded angles through Equation (16) with the old gain reduces the measured understeer gradient from $K_{\text{us}} = 0.012$ to $K_{\text{us}} = 0.0092$ rad/(m/s^2) and yields the updated cornering stiffnesses in Equation (12).

1.10 Parameter Summary

Table 6 compares all measured values with the F1/10th gym simulation defaults.

Table 6: Identified vehicle model parameters.

Parameter	Symbol	Measured	Sim default	Unit	Status
Wheelbase	L	0.324	0.324	m	Measured
Friction coeff.	μ	0.745	1.049	–	Tested
Front corn. stiff.	$C_{\alpha f}$	53.6	4.718	$\mathrm{N\,rad^{-1}}$	Tested[a]
Rear corn. stiff.	$C_{\alpha r}$	44.6	5.456	$\mathrm{N\,rad^{-1}}$	Tested[a]
Understeer gradient	K_{us}	0.0092	–	$\mathrm{rad/(m/s^2)}$	Tested
Rolling resistance	c_{roll}	0.163	0.003	–	Tested[b]
Effective K_t	$K_{t,\mathrm{eff}}$	0.583	–	$\mathrm{N\,A^{-1}}$	Tested
Max drive torque	T_{max}	1.93	22.9	$\mathrm{N\,m}$	Tested[c]
Max brake decel.	a_{brake}	7.3	10.0	$\mathrm{m\,s^{-2}}$	Tested
Max velocity	v_{max}	5.17	20.0	$\mathrm{m\,s^{-1}}$	Tested
Max acceleration	a_{max}	7.3	9.51	$\mathrm{m\,s^{-2}}$	Tested
Max steer. rate	$\dot{\delta}_{\mathrm{max}}$	2.85	3.2	$\mathrm{rad\,s^{-1}}$	Tested
Servo gain ratio	$\delta_{\mathrm{act}}/\delta_{\mathrm{cmd}}$	1.07	1.0	–	Calibrated

[a]Combined K_{us} + yaw-frequency method (Section 1.3); corrected for nonlinear servo response (Section 1.9). Absolute values depend on assumed ω_n. The ratio $C_{\alpha f}/C_{\alpha r} = 1.20$ is better constrained.
[b]Includes both tire rolling resistance and drivetrain friction; measured via coast-down test (Section 1.4).
[c]At the wheel ($T = F_{\mathrm{max}} \times r_{\mathrm{eff}}$), at the VESC current limit of 65 A.

Discussion. The measured parameters differ substantially from the simulation defaults:

- **Friction**: $\mu = 0.75$ vs. 1.05 in simulation. The real car has about 29% less grip.
- **Maximum velocity**: $v_{\mathrm{max}} = 5.17\,\mathrm{m\,s^{-1}}$ vs. $20\,\mathrm{m\,s^{-1}}$ in simulation. The real vehicle's speed envelope is roughly 4× smaller.
- **Motor torque** ($K_{t,\mathrm{eff}} = 0.583\,\mathrm{N\,A^{-1}}$): 8% lower than the theoretical $K_t\,G/r_{\mathrm{eff}} = 0.632\,\mathrm{N\,A^{-1}}$, indicating drivetrain losses.
- **Rolling resistance**: $c_{\mathrm{roll}} = 0.163$ is significantly higher than the simulation default of 0.003, reflecting the combined tire and drivetrain friction of a geared 4×4 chassis.
- **Understeer**: the vehicle understeers with $K_{\mathrm{us}} = 0.0092$, requiring progressively more steering input at higher lateral accelerations.

These measurements confirm that the simulation defaults are not representative of the physical vehicle and must be replaced before deploying the MPC controller on the real car.